SOCIOL 723: Social Statistics II

Week 1 (Thursday), OLS in Matrix Form

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Today

The Matrix Form and Its Geometry

Partialling Out in General

Tuesday built everything in scalar form. Today: the same objects again in matrix form, for general understanding, followed by lab.

★ = essential, you must understand this ★ = for understanding, not examined

Matrix Form

Why Bother

- With k regressors, scalar algebra needs k normal equations and becomes unreadable. Matrix algebra stays two lines long for any k.
- The geometry, projection onto a subspace, is invisible in scalar form and obvious in matrix form.
- In the literature, most estimators you will meet (fixed effects, IV, 2SLS, ridge) are *written* as small variations on $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$. We will keep working in scalars, but recognizing this form is what lets you read modern papers.

But

Nothing in this section is new content. It is the *same* normal equations you derived on Tuesday, written more compactly. The lab that follows practices every piece of linear algebra we use.

Vectors and Matrices, From Zero ★

No linear algebra is assumed in this course. Here is the part we need, from nothing.

- A **vector** is a list of numbers written in a column. X_i is the list describing unit i: one entry per variable, with a 1 on top for the intercept.
- A **matrix** is a rectangular table of numbers: an $n \times k$ matrix has n rows and k columns.
- The **transpose** $'$ tips a column onto its side: if X_i is a $k \times 1$ column, then X_i' is the same list lying down, $1 \times k$.

The one formula to internalize

$$X_i'\boldsymbol{\beta} = \beta_0 \cdot 1 + \beta_1 X_{i1} + \cdots + \beta_{k-1} X_{i,k-1} :$$

a row times a column is a *single number*: multiply matching entries, then add. So $X_i'\boldsymbol{\beta}$ is nothing but unit i's fitted value, written compactly. Whenever you see $X_i'\boldsymbol{\beta}$, read it as "the regression line evaluated at unit i."

Multiplying and Inverting

The only two operations we use. **Multiplication** extends the row-times-column rule: entry (r, c) of $\boldsymbol{AB}$ is (row r of $\boldsymbol{A}$) times (column c of $\boldsymbol{B}$):

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Dimensions must match, $(n \times k)(k \times 1) = (n \times 1)$, and order matters: $\boldsymbol{AB} \neq \boldsymbol{BA}$ in general.

Inversion is the matrix version of dividing. For a number, $a^{-1}a = 1$; for a square matrix, $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$, the *identity* matrix (ones on the diagonal, zeros elsewhere), which leaves whatever it multiplies unchanged.

- Just as $1/a$ fails to exist when $a = 0$, $\boldsymbol{A}^{-1}$ fails to exist when the matrix collapses information; that is the "rank" condition, explained shortly.
- You will never invert a matrix by hand here; `R`'s `solve()` does it. What matters is knowing what the symbol says.

Stacking the Data

For n units and k regressors (including the constant):

$$\boldsymbol{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1}, \quad \boldsymbol{X} = \begin{bmatrix} X_1' \\ \vdots \\ X_n' \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1,k-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,k-1} \end{bmatrix}_{n\times k}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model, stacked over units:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \qquad \Longleftrightarrow \qquad Y_i = X_i'\boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n.$$

Two objects do all the work: $\boldsymbol{X}'\boldsymbol{X} = \sum_i X_i X_i'$ (a $k \times k$ matrix) and $\boldsymbol{X}'\boldsymbol{y} = \sum_i X_i Y_i$ (a $k \times 1$ vector).

Least Squares in Matrix Form: the Expansion

The sum of squared errors is the error vector times itself:

$$S(b) = \sum_i (Y_i - X_i' b)^2 = (\boldsymbol{y} - \boldsymbol{X} b)'(\boldsymbol{y} - \boldsymbol{X} b).$$

Two transpose rules, used constantly

$(\boldsymbol{A} + \boldsymbol{B})' = \boldsymbol{A}' + \boldsymbol{B}'$, and $(\boldsymbol{A}\boldsymbol{B})' = \boldsymbol{B}'\boldsymbol{A}'$: transposing a product *reverses the order*. Hence $(\boldsymbol{X} b)' = b'\boldsymbol{X}'$.

Multiply out, term by term:

$$S(b) = \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X} b - b'\boldsymbol{X}'\boldsymbol{y} + b'\boldsymbol{X}'\boldsymbol{X} b = \boldsymbol{y}'\boldsymbol{y} - 2\, b'\boldsymbol{X}'\boldsymbol{y} + b'\boldsymbol{X}'\boldsymbol{X} b.$$

- Why do the middle terms merge? Each is 1×1, a single number, and a number equals its own transpose: $(b'\boldsymbol{X}'\boldsymbol{y})' = \boldsymbol{y}'\boldsymbol{X} b$. The two terms are the *same* number written twice.

Minimizing: Two Matrix Derivatives, From Zero

$\partial S/\partial b$ just stacks the k partials into a column; two stock rules cover everything:

The rules, with their scalar shadows

$$\frac{\partial(a'b)}{\partial b} = a \qquad \text{since } a'b = \textstyle\sum_j a_j b_j \text{ gives } \partial/\partial b_j = a_j \qquad (\text{cf. } \tfrac{d}{db}\, ab = a)$$

$$\frac{\partial(b'\boldsymbol{A}b)}{\partial b} = 2\boldsymbol{A}b \qquad \text{for symmetric } \boldsymbol{A} \qquad (\text{cf. } \tfrac{d}{db}\, ab^2 = 2ab)$$

Apply them with $a = \boldsymbol{X}'\boldsymbol{y}$ and $\boldsymbol{A} = \boldsymbol{X}'\boldsymbol{X}$ (symmetric):

$$\frac{\partial S(b)}{\partial b} = -2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}b \overset{!}{=} 0 \quad \Longrightarrow \quad \underbrace{\boldsymbol{X}'\boldsymbol{X}\,\hat{\beta} = \boldsymbol{X}'\boldsymbol{y}}_{\text{normal equations}}.$$

The *same* k equations you wrote out one at a time on Tuesday. Why a minimum? $v'\boldsymbol{X}'\boldsymbol{X}v = \|\boldsymbol{X}v\|^2 \geq 0$ for all v: S is a bowl. (Shorthand: $\boldsymbol{X}'\boldsymbol{X} \succeq 0$, the matrix $\geq$.)

The Normal Equations, Slowly: the Bivariate Case

Take $k = 2$ with $X_i = (1, X_i)'$, so $b = (b_0, b_1)'$. The two building blocks are

$$\boldsymbol{X}'\boldsymbol{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Write out $\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}'\boldsymbol{y}$ one row at a time:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i X_i = \sum_i Y_i \iff \sum_i \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right) = 0$$

$$\hat{\beta}_0 \sum_i X_i + \hat{\beta}_1 \sum_i X_i^2 = \sum_i X_i Y_i \iff \sum_i X_i\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i\right) = 0$$

- Row by row, the matrix equation *is* Tuesday's (N1) and (N2): residuals sum to zero, and residuals are orthogonal to the regressor.
- One matrix line, k scalar equations. That is the entire content of the compact notation: nothing new, everything stacked.

The Estimator, Existence, and Rank

What "rank" means, in words

The rank of $\boldsymbol{X}$ is the number of genuinely distinct columns it contains: columns that cannot be rebuilt as combinations of the others. *Full column rank*, $\text{rank}(\boldsymbol{X}) = k$, says no column is redundant: the data carry k separate pieces of information, one per coefficient.

The OLS estimator

If $\text{rank}(\boldsymbol{X}) = k$, $\boldsymbol{X}'\boldsymbol{X}$ is invertible and

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \left(\mathbb{E}_n[X_i X_i']\right)^{-1}\mathbb{E}_n[X_i Y_i].$$

- Full rank needs $n \geq k$; it is the matrix version of "X must vary."
- Failures you will actually meet: the *dummy variable trap*; including a variable and a rescaling of it; a group indicator constant within a fixed effect.
- If rank fails, `R` silently drops a column and reports `NA`. That `NA` is never a mystery.

Sanity Check: Recovering the Scalar Formula ★

Let $k = 2$ with $X_i = (1, X_i)'$. Then

$$\boldsymbol{X}'\boldsymbol{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Using $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$,

$$\hat{\beta}_1 = \frac{n \sum_i X_i Y_i - \sum_i X_i \sum_i Y_i}{n \sum_i X_i^2 - (\sum_i X_i)^2} = \frac{\mathrm{Cov}_n(X,Y)}{\mathrm{Var}_n(X)}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Exactly what we derived by hand. The matrix form is a change of language, not of substance.

Why Projection? It Is Tuesday's Argmin, Re-Read ★

Tuesday defined OLS by a minimization: $\hat{\beta} = \arg\min_b \sum_{i=1}^n (Y_i - X_i'b)^2$.

The one observation everything hangs on

A sum of squared entries is a squared *length*: $\sum_i (Y_i - X_i'b)^2 = \|\boldsymbol{y} - \boldsymbol{X}b\|^2$, the squared *distance* between two points of $\mathbb{R}^n$, the data $\boldsymbol{y}$ and the candidate fit $\boldsymbol{X}b$. So the argmin says, word for word: among all fits the model can produce, pick the one closest to the data.

- The candidate fits $\boldsymbol{X}b$ form a flat sheet, a "plane" (next frame), and "closest point on a plane" is a problem geometry solved long before calculus: the foot of the perpendicular.
- The right angle *is* the first-order condition: perpendicular to every column of $\boldsymbol{X}$ means $\boldsymbol{X}'\hat{\boldsymbol{e}} = \boldsymbol{0}$, the normal equations, with no differentiation anywhere.
- Nothing new is computed: projection is the same argmin, *answered by geometry instead of calculus*. Next: the smallest possible case.

What Does "All the Fits" Look Like? ★

Read matrix-times-vector as a recipe for mixing columns:

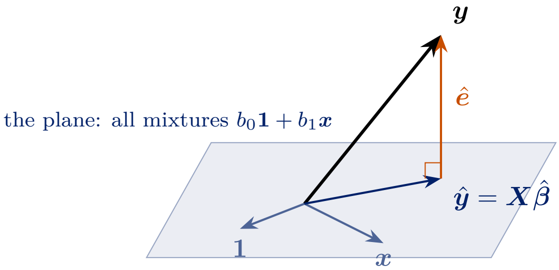
$$\boldsymbol{X}b = b_1 \cdot (\text{column } 1) + b_2 \cdot (\text{column } 2) + \cdots + b_k \cdot (\text{column } k) :$$

each coefficient pours in that much of its column. The set of candidate fits is therefore *every possible mixture of the k columns*.

- **One column** (intercept only): the mixtures are all multiples of $\mathbf{1}$, a line through the origin. Projecting $\boldsymbol{y}$ onto that line gives $\hat{b} = \bar{Y}$: the sample mean is a projection.
- **Two columns**: the mixtures of two arrows fill a flat plane through the origin.
- k **columns**: the mixtures fill a k-dimensional flat subspace of $\mathbb{R}^n$. "k-dimensional" simply counts the free coefficients $b_1, \ldots, b_k$.
- $\boldsymbol{y}$ generally sits *off* this subspace; the gap is the residual. A column that is itself a mixture of the others adds no new direction, so the subspace has dimension below k: rank, again.

The Picture: $n = 3$, Two Columns ★

Three observations ($n = 3$), two columns: $\boldsymbol{X} = [\,\mathbf{1}\ \ \boldsymbol{x}\,]$ is 3×2. The mixtures $b_0\mathbf{1} + b_1\boldsymbol{x}$ fill the shaded plane; the data $\boldsymbol{y}$ floats *off* it, in the third dimension.



- OLS drops the perpendicular from $\boldsymbol{y}$ to the plane: the foot is the fit $\hat{\boldsymbol{y}}$, the gap is the residual $\hat{\boldsymbol{e}}$. Perpendicular to the plane = perpendicular to *both* columns: $\boldsymbol{X}'\hat{\boldsymbol{e}} = \mathbf{0}$, the normal equations.
- Fitting is dropping a perpendicular. Everything else is bookkeeping.

OLS Is a Projection ★

Where does the famous "hat matrix" come from? Pure substitution. We already have $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$, and the fitted vector is $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\beta}$:

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\beta} = \underbrace{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'}_{\boldsymbol{P}}\,\boldsymbol{y}, \qquad \hat{\boldsymbol{e}} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \underbrace{(\boldsymbol{I}_n - \boldsymbol{P})}_{\boldsymbol{M}}\,\boldsymbol{y}.$$

- $\boldsymbol{P}$, the hat matrix (it puts the hat on $\boldsymbol{y}$), is simply whatever multiplies the data to produce the fit: one matrix that drops the perpendicular onto the plane $\text{col}(\boldsymbol{X}) = \{\boldsymbol{X}b\}$.
- $\boldsymbol{M}$, the residual maker, is whatever is left: it carries the data straight to the residual.
- Nothing new was derived. $\boldsymbol{P}$ and $\boldsymbol{M}$ are Tuesday's estimator, repackaged as machines acting on $\boldsymbol{y}$.

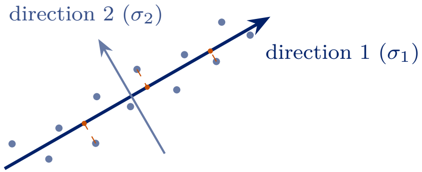
$\boldsymbol{P}$, $\boldsymbol{M}$, and Leverage: the Facts We Actually Use ★

1. **Projecting twice changes nothing**: $\boldsymbol{PP} = \boldsymbol{P}$ and $\boldsymbol{MM} = \boldsymbol{M}$, and both are symmetric. Once you are on the plane, projecting again leaves you where you stand.
2. **Traces count dimensions**: $\text{tr}(\boldsymbol{P}) = k$, $\text{tr}(\boldsymbol{M}) = n - k$. This redeems Tuesday's IOU: $E[\sum_i \hat{e}_i^2] = \sigma^2(n-k)$, the $n-k$ inside s^2.
3. **Leverage**: the ith diagonal entry of $\boldsymbol{P}$, $h_{ii} = X_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}X_i = \partial\hat{Y}_i/\partial Y_i$, is how hard observation i's own outcome pulls its own fitted value. $\sum_i h_{ii} = k$; high leverage means unusual *covariates*; and $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$: high-leverage points drag the line toward themselves. Returns next week in HC2/HC3.

For the traces: $\text{tr}(\boldsymbol{P}) = \text{tr}\left((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\right) = \text{tr}(\boldsymbol{I}_k) = k$, using the cyclic property $\text{tr}(ABC) = \text{tr}(BCA)$.

The Same Idea, Run in Reverse: the SVD ★

In OLS the subspace was *handed to us*: $\boldsymbol{X}$'s columns. Flip the question: which subspace would the data choose for themselves?



The dashed drops are perpendiculars again: the best one-dimensional summary of the cloud is the line whose perpendicular losses are smallest.

- A data cloud has a *long* direction and a *short* one. The singular value decomposition, $\boldsymbol{X} = \boldsymbol{U\Sigma V'}$, finds every direction and measures each one's share of the data: $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. The rank from earlier is how many are positive.
- **Keep the top r directions** = project every observation onto the best r-dimensional subspace: the closest rank-r version of the data, by the same least-squares logic as OLS.
- In OLS you project onto a *given* plane; the SVD *finds* the plane worth projecting onto.

Latent Semantic Analysis: Projection Applied to Text ★

Let $\boldsymbol{X}$ be a document–term matrix: one row per document, one column per word, entries counting occurrences. Thousands of columns; far fewer *themes*.

- Truncate the SVD at $r \approx 100$ directions: each document becomes a mixture of r latent semantic dimensions, and words that co-occur across documents get pulled onto the same dimension. That is *latent semantic analysis* (Deerwester et al. 1990).
- Sociology runs on this geometry: Kozlowski, Taddy, and Evans (2019) *project* word vectors onto cultural axes such as rich–poor to track the meaning of class across a century, and the operation is literally the projection formula from a few frames back.
- Course connections: ridge regression shrinks along exactly these singular directions (Week 12), and synthetic control's factor model is a low-rank story about a panel (Week 11).

None of this is examinable: it is where the geometry goes when you point it at data sociologists care about.

Unbiasedness, in Matrix Form ★

Tuesday's two-line scalar proof, now for any k:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\epsilon},$$

$$E[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\underbrace{E[\boldsymbol{\epsilon} \mid \boldsymbol{X}]}_{=\mathbf{0}} = \boldsymbol{\beta}. \qquad \blacksquare$$

- Note where the work is done: *only* by A4.
- Heteroskedasticity, clustering, and non-normality do not bias $\hat{\boldsymbol{\beta}}$. They affect its *variance*, next week.
- A4 fails whenever an omitted variable, reverse causation, measurement error in X, or selection puts something correlated with X into ϵ. That is the entire causal-inference problem in one sentence.

The Gauss–Markov Theorem ★

Theorem

Under A1–A5, $\hat{\boldsymbol{\beta}}$ is the Best Linear Unbiased Estimator: among all $\tilde{\boldsymbol{\beta}} = \boldsymbol{C}\boldsymbol{y}$ with $\boldsymbol{C}$ a function of $\boldsymbol{X}$ and $E[\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta}$, $\mathrm{Var}(\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}) - \mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) \succeq 0$.

The Gauss–Markov Theorem ★

Theorem

Under A1–A5, $\hat{\beta}$ is the Best Linear Unbiased Estimator: among all $\tilde{\beta} = \boldsymbol{C}\boldsymbol{y}$ with $\boldsymbol{C}$ a function of $\boldsymbol{X}$ and $E[\tilde{\beta} \mid \boldsymbol{X}] = \beta$, $\mathrm{Var}(\tilde{\beta} \mid \boldsymbol{X}) - \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}) \succeq 0$.

Proof. Write $\boldsymbol{C} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{D}$. Unbiasedness for all β forces $\boldsymbol{D}\boldsymbol{X} = \mathbf{0}$. Then

$$
\begin{aligned}
\mathrm{Var}(\boldsymbol{C}\boldsymbol{y} \mid \boldsymbol{X}) &= \sigma^2\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{D}\right]\left[\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} + \boldsymbol{D}'\right] \\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} + \sigma^2\underbrace{(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{D}'}_{=\mathbf{0}} + \sigma^2\underbrace{\boldsymbol{D}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}}_{=\mathbf{0}} + \sigma^2\boldsymbol{D}\boldsymbol{D}' \\
&= \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}) + \sigma^2\boldsymbol{D}\boldsymbol{D}' \succeq \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}). \qquad \blacksquare
\end{aligned}
$$

Partialling Out

The Frisch–Waugh–Lovell Theorem ★

Partition $\boldsymbol{X} = [\,\boldsymbol{X}_1 \quad \boldsymbol{X}_2\,]$: a treatment of interest and controls. Let $\boldsymbol{M}_2 = \boldsymbol{I} - \boldsymbol{X}_2(\boldsymbol{X}_2'\boldsymbol{X}_2)^{-1}\boldsymbol{X}_2'$ and define

$$\tilde{\boldsymbol{X}}_1 = \boldsymbol{M}_2\boldsymbol{X}_1, \qquad \tilde{\boldsymbol{y}} = \boldsymbol{M}_2\boldsymbol{y}.$$

Theorem

The OLS estimate $\hat{\boldsymbol{\beta}}_1$ from the full regression equals

$$\hat{\boldsymbol{\beta}}_1 = (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{y}} = (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\boldsymbol{y},$$

and the residuals from the short regression are numerically identical to $\hat{\boldsymbol{e}}$ from the full one.

This is the three-step recipe from earlier, now for any number of controls.

Proof of FWL ★

Premultiply the model by $\boldsymbol{M}_2$ and use $\boldsymbol{M}_2\boldsymbol{X}_2 = \mathbf{0}$:

$$\boldsymbol{M}_2\boldsymbol{y} = \boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{\beta}_1 + \underbrace{\boldsymbol{M}_2\boldsymbol{X}_2}_{=\mathbf{0}}\boldsymbol{\beta}_2 + \boldsymbol{M}_2\boldsymbol{\epsilon} \quad \Longrightarrow \quad \tilde{\boldsymbol{y}} = \tilde{\boldsymbol{X}}_1\boldsymbol{\beta}_1 + \boldsymbol{M}_2\boldsymbol{\epsilon}.$$

Proof of FWL ★

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$$\boldsymbol{M}_2\boldsymbol{y} = \boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{\beta}_1 + \underbrace{\boldsymbol{M}_2\boldsymbol{X}_2}_{=\mathbf{0}}\boldsymbol{\beta}_2 + \boldsymbol{M}_2\boldsymbol{\epsilon} \quad\Longrightarrow\quad \tilde{\boldsymbol{y}} = \tilde{\boldsymbol{X}}_1\boldsymbol{\beta}_1 + \boldsymbol{M}_2\boldsymbol{\epsilon}.$$

For the estimator, decompose the fitted model $\boldsymbol{y} = \boldsymbol{X}_1\hat{\boldsymbol{\beta}}_1 + \boldsymbol{X}_2\hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{e}}$ and premultiply by $\tilde{\boldsymbol{X}}_1' = \boldsymbol{X}_1'\boldsymbol{M}_2$:

$$\tilde{\boldsymbol{X}}_1'\boldsymbol{y} = \tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1\hat{\boldsymbol{\beta}}_1 + \boldsymbol{X}_1'\underbrace{\boldsymbol{M}_2\boldsymbol{X}_2}_{=\mathbf{0}}\hat{\boldsymbol{\beta}}_2 + \underbrace{\boldsymbol{X}_1'\boldsymbol{M}_2\hat{\boldsymbol{e}}}_{=0},$$

using $\boldsymbol{M}_2\hat{\boldsymbol{e}} = \hat{\boldsymbol{e}}$ (since $\boldsymbol{X}_2'\hat{\boldsymbol{e}} = \mathbf{0}$) and $\boldsymbol{X}_1'\hat{\boldsymbol{e}} = \mathbf{0}$. Solving gives the result. Using $\tilde{\boldsymbol{y}}$ instead of $\boldsymbol{y}$ changes nothing because $\tilde{\boldsymbol{X}}_1'\boldsymbol{M}_2 = \tilde{\boldsymbol{X}}_1'$. ■

Why FWL Matters More Than It Looks

1. **Interpretation.** "Controlling for W" = discarding all covariance between X_1 and W. Your estimate lives on what remains.
2. **Computation.** Fixed-effects estimators *are* FWL: regressing on unit dummies is the same as demeaning within unit (Week 10).
3. **Diagnostics.** The added-variable plot is the only two-dimensional picture that faithfully represents a multivariate slope.
4. **Generality.** Nothing in the argument requires $\boldsymbol{X}_2$ to enter linearly, it only requires that we can compute the part of X_1 that $\boldsymbol{X}_2$ predicts. That observation is the seed of a large literature, and we return to it briefly in Week 13.

Omitted Variables, in Matrix Form ★

With $\boldsymbol{X}_1$ included and $\boldsymbol{X}_2$ omitted,

$$E[\tilde{\boldsymbol{\beta}}_1 \mid \boldsymbol{X}] = \boldsymbol{\beta}_1 + \underbrace{(\boldsymbol{X}_1'\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1'\boldsymbol{X}_2}_{\hat{\boldsymbol{\delta}}}\,\boldsymbol{\beta}_2.$$

- Each column of $\hat{\boldsymbol{\delta}}$ is the coefficient vector from regressing one omitted variable on all the included ones, exactly δ_1 from the scalar case, stacked.
- With several omitted variables the bias is a *sum* of terms which may offset, so "the bias must be positive" arguments require care.
- Sensitivity analysis (Week 6) formalizes the question: how strong would an omitted confounder have to be to overturn the result?

Where This Leaves Us

- OLS minimizes squared errors. Everything else, the slope formula, the normal equations, FWL, OVB, follows from that one criterion by algebra.
- In the population, OLS estimates the best linear approximation to the CEF. Under A1–A4 the sample version is unbiased for it.
- Whether that coefficient is a *causal effect* is a separate question that regression cannot answer.
- **Next week**: we have $\hat{\beta}$; now we need to know how uncertain it is, and the standard error your software prints is usually the wrong one.

This week's reading

MHE Chapter 3; CCI Chapter 6. The lab practices the matrix algebra hands-on and computes OLS by hand in `R`.

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